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CBSE CLASS 10 | MATHEMATICS

# QUADRATIC EQUATIONS

Master Concept, Formula & Visual Problem-Solving Handbook

**Academic Year 2026–27**

**Structure the equation. Inspect the discriminant.**
Solve with absolute precision.

Standard Form | Factorization | Sreedharacharya Formula | Nature of Roots | Kinematics

A student-first Maths Manthan conceptual and visual revision resource



## What This Handbook Includes

- Standard form criteria and parabolic geometry of roots
- Splitting middle term with rational and radical surds
- Sreedharacharya derivation of the Quadratic Formula
- Tri-state discriminant analysis ($D > 0, D = 0, D < 0$)
- High-frequency parameter $k$ equal-roots protocols
- Complete word problem blueprints: streams, taps, work

## Pedagogical Strategy: Foundations First

A quadratic equation represents the balance of squared relationships. Roots are not mere symbols; geometrically, they are the points where the trajectory crosses ground level ($y = 0$).

$$\mathbf{y = ax^2 + bx + c = 0} \iff \text{Roots on } \mathbf{x}\text{-axis}$$

$$\mathbf{D = b^2 - 4ac} \iff \text{Nature of Roots}$$

Always structure equations into standard form before applying formulas!

## CBSE 2026–27 Curriculum Verification & Rationalized Scope

**Official Scope (Prescribed & Tested):** Standard form of a quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$). Solutions of quadratic equations (only real roots) by factorization and by using the quadratic formula. Relationship between discriminant and nature of roots. Situational word problems arising from daily activities.

**Excluded / Deleted Topics (Non-Examinable):**

- **"Completing the Square" Method:** Completely deleted from the NCERT syllabus.
- **Complex / Imaginary Roots:** When $D < 0$, students must simply state "No real roots exist".
- **Equations Reducible to Quadratic Form:** Complex multi-nested substitutions are deleted.

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Prepared for learning and conceptual mastery – Strictly aligned with CBSE Class 10 Syllabus 2026–27.

CHAPTER 04

CORE SYLLABUS

# QUADRATIC EQUATIONS

Module 1 | Foundations: Standard Form, Degree & Physical Origins

## 1.1 Definition and the Critical Leading Coefficient Condition

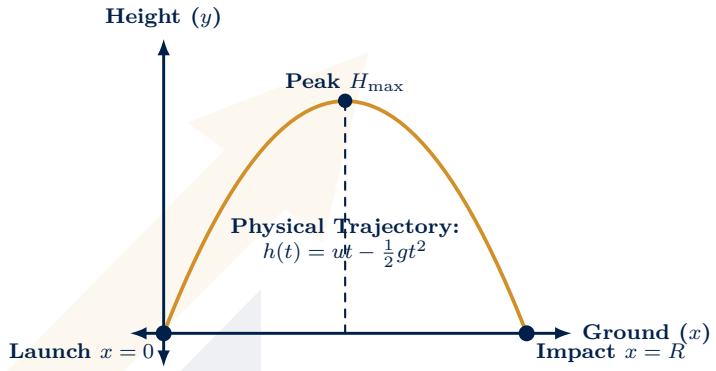
A quadratic equation in the single variable $x$ is an algebraic equation of second degree.

**The Standard Form of a Quadratic Equation:**

$$\mathbf{ax^2 + bx + c = 0}$$

where $a, b, c \in \mathbb{R}$ are real numbers, and $\mathbf{a \neq 0}$.

- **Leading Coefficient ($a$):** The multiplier of $x^2$. If $a = 0$, the equation degenerates into $bx + c = 0$, which is **linear**, not quadratic!
- **Linear Coefficient ($b$):** Can be zero (e.g., $ax^2 + c = 0$ is a pure quadratic equation).
- **Constant Term ($c$):** Can be zero (e.g., $ax^2 + bx = 0 \implies x(ax + b) = 0$).




## 1.2 Testing Whether an Algebraic Expression is Quadratic

TRAP ALERT

**Never Judge an Equation Before Full Expansion!** An equation cannot be classified by visual inspection alone. You must expand all products, clear denominators, and gather all terms to one side:

- Example A: $(x-2)^2 + 1 = 2x - 3 \implies x^2 - 4x + 4 + 1 = 2x - 3 \implies \mathbf{x^2 - 6x + 8 = 0}$ **(Quadratic!)**
- Example B: $x(x+1) + 8 = (x+2)(x-2) \implies x^2 + x + 8 = x^2 - 4 \implies \mathbf{x + 12 = 0}$ **(Linear, NOT Quadratic!)**
- Example C: $x + \frac{1}{x} = x^2 \implies x^2 + 1 = x^3 \implies \mathbf{x^3 - x^2 - 1 = 0}$ **(Cubic, NOT Quadratic!)**

### Worked Archetype 1 | Algebraic Simplification & Standard Form Verification

**Problem:** Check whether the equation $(x+2)^3 = 2x(x^2 - 1)$ is a quadratic equation. If so, write it in standard form.

**Step 1: Expand LHS using algebraic identity** $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$**:**

$$\text{LHS} = (x+2)^3 = x^3 + 3(x^2)(2) + 3(x)(2^2) + 2^3 = x^3 + 6x^2 + 12x + 8$$

**Step 2: Expand RHS:** $\text{RHS} = 2x(x^2 - 1) = 2x^3 - 2x$.

**Step 3: Equate LHS and RHS and transfer all terms to one side:**

$$x^3 + 6x^2 + 12x + 8 = 2x^3 - 2x \implies 2x^3 - x^3 - 6x^2 - 2x - 12x - 8 = 0$$

$$\mathbf{x^3 - 6x^2 - 14x - 8 = 0}$$

**Conclusion:** Since the highest power of $x$ is 3 (degree is 3), this is a **cubic equation, NOT a quadratic equation**.

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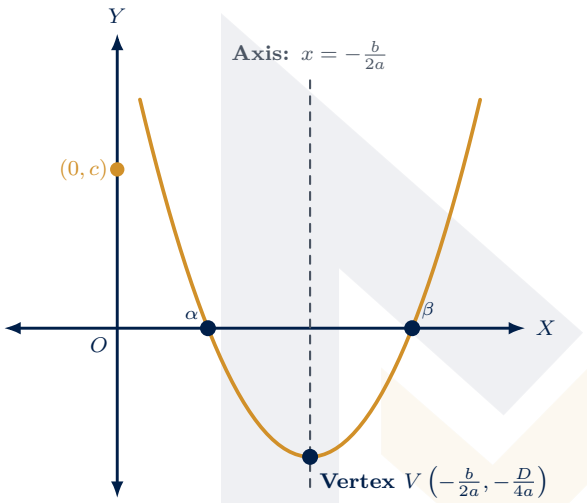
# QUADRATIC EQUATIONS

Module 2 | Parabolic Geometry: Roots, Intercepts & Symmetry

## 2.1 Geometric Meaning of Roots: $x$-Intercepts of a Parabola

A real number $\alpha$ is called a **root** (or solution/zero) of the quadratic equation $ax^2 + bx + c = 0$ if and only if $a\alpha^2 + b\alpha + c = 0$.

- Geometrically, the roots are the **$x$-coordinates of the points where the parabola $y = ax^2 + bx + c$ intersects the horizontal $x$-axis**.
- Since a parabola can intersect a line in at most two points, a quadratic equation can have **at most two real roots**.




**Anatomy of the Quadratic Function $y = ax^2 + bx + c$:**

- **Curvature Direction:**
  - If $a > 0$: Opens upwards ($\cup$, concave up, has a minimum).
  - If $a < 0$: Opens downwards ($\cap$, concave down, has a maximum).
- **Axis of Symmetry:** Passes through the exact midpoint of roots:

$$\mathbf{x = \frac{\alpha + \beta}{2} = -\frac{b}{2a}}$$

- **Vertex Coordinates:** Extreme point of parabola:

$$\mathbf{V\left(-\frac{b}{2a}, -\frac{D}{4a}\right)} \quad \text{where } D = b^2 - 4ac$$

- **$y$-Intercept:** Set $x = 0 \implies \mathbf{y = c}$.

## 2.2 The Zero-Product Principle

**Fundamental Algebraic Zero-Product Axiom:**

$$\mathbf{A \cdot B = 0 \iff A = 0} \quad \text{or} \quad \mathbf{B = 0} \quad \text{(or both)}$$

If a quadratic equation is factorized into linear factors $(x - \alpha)(x - \beta) = 0$, the roots are $\mathbf{x = \alpha}$ and $\mathbf{x = \beta}$.

### Worked Archetype 2 | Verifying Roots of a Quadratic Equation

**Problem:** Determine whether $x = -1$ and $x = \frac{1}{2}$ are solutions of the equation $2x^2 - x - 1 = 0$.

**Check for $x = -1$:** Substitute $x = -1$ into LHS:

$$\text{LHS} = 2(-1)^2 - (-1) - 1 = 2(1) + 1 - 1 = 2 \neq 0 \implies \mathbf{x = -1} \text{ is NOT a root.}$$

**Check for $x = 1$:** Substitute $x = 1$ into LHS:

$$\text{LHS} = 2(1)^2 - (1) - 1 = 2 - 1 - 1 = 0 = \text{RHS} \implies \mathbf{x = 1} \text{ IS a root.}$$

**Check for $x = -\frac{1}{2}$:** $\text{LHS} = 2\left(-\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right) - 1 = 2\left(\frac{1}{4}\right) + \frac{1}{2} - 1 = \frac{1}{2} + \frac{1}{2} - 1 = 0 \implies \mathbf{x = -\frac{1}{2}}$ IS a root.

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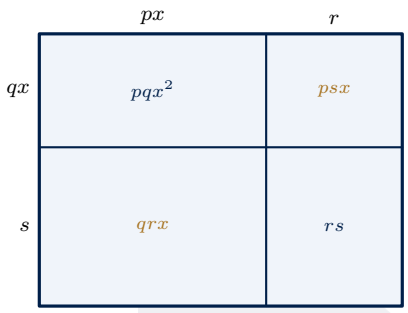
# QUADRATIC EQUATIONS

Module 3 | Factoring by Splitting the Middle Term & Radical Surds

## 3.1 The AC-Method Protocol for Factoring $ax^2 + bx + c = 0$

To factor $ax^2 + bx + c = 0$, we decompose the middle term $bx$ into two terms $px$ and $qx$ such that:

$$\mathbf{p + q = b} \quad \text{and} \quad \mathbf{p \cdot q = a \cdot c}$$




**Total Area** $= (px + r)(qx + s)$

**Step-by-Step Factoring Protocol:**

1. Multiply $a$ and $c$ to form the product $P = ac$.
2. List pairs of factors of $P$ that sum up to $b$.
3. Split $bx$ as $px + qx$.
4. Factor by grouping terms in pairs:
$$ax^2 + px + qx + c = 0$$
5. Extract common binomial factor:
$$(px + r)(qx + s) = 0$$
6. Equate each linear factor to zero to find roots.

## 3.2 Factoring with Radical (Surd) Coefficients

TRAP ALERT

**Handling Radicals in Middle and Constant Terms:** When equations involve square roots like $\sqrt{k}$, calculate $ac$ carefully:

- For $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$: Product $ac = (\sqrt{2})(5\sqrt{2}) = 5 \times 2 = \mathbf{10}$.
- Find two numbers whose product is 10 and sum is 7: 5 and 2 ($5 \times 2 = 10, 5 + 2 = 7$).
- Rewrite $2x$ strategically as $\sqrt{2} \cdot \sqrt{2}x$ to enable grouping!

### Worked Archetype 3 | Factorization with Radical Coefficients

**Problem:** Solve the quadratic equation by factorization: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$.

**Step 1: Compute product $ac$ and target sum $b$:**

$$a = \sqrt{2}, \quad b = 7, \quad c = 5\sqrt{2} \implies ac = \sqrt{2} \times 5\sqrt{2} = 10$$

We need factors of 10 that add up to 7. The factors are 5 and 2.

**Step 2: Split the middle term $7x$ into $5x + 2x$:**

$$\sqrt{2}x^2 + 2x + 5x + 5\sqrt{2} = 0$$

**Step 3: Group terms in pairs and factor out common elements:** Notice that $2 = (\sqrt{2})^2$:

$$\sqrt{2}x(x + \sqrt{2}) + 5(x + \sqrt{2}) = 0$$

**Step 4: Extract common binomial factor $(x + \sqrt{2})$:**

$$(x + \sqrt{2})(\sqrt{2}x + 5) = 0$$

**Step 5: Apply the Zero-Product Principle:**

$$x + \sqrt{2} = 0 \implies \mathbf{x = -\sqrt{2}} \quad \text{or} \quad \sqrt{2}x + 5 = 0 \implies \mathbf{x = -\frac{5}{\sqrt{2}} = -\frac{5\sqrt{2}}{2}}$$

**Final Statement:** The roots of the equation are $\mathbf{-\sqrt{2}}$ and $\mathbf{-\frac{5\sqrt{2}}{2}}$.

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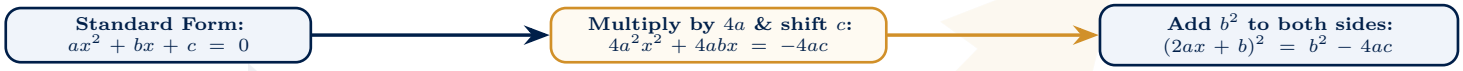
CORE SYLLABUS

# QUADRATIC EQUATIONS

Module 4 | The Quadratic Formula: Sreedharacharya's Classical Derivation

## 4.1 Historical Context: Sreedharacharya's Elegant Technique

Ancient Indian mathematician **Sreedharacharya** (c. 8th–9th century CE) derived the universal quadratic formula using a brilliant device: instead of dividing by $a$ (which introduces messy fractions), he **multiplied the entire equation by** $4a$. This creates an exact square term $(2ax)^2$ and $4abx$, which perfectly completes the binomial square $(2ax+b)^2$!




## 4.2 Rigorous Step-by-Step Proof of the Quadratic Formula

**Algebraic Derivation of the Formula:**

1. Start with the standard quadratic equation:
$$ax^2 + bx + c = 0 \quad (a \neq 0)$$
2. Transpose the constant term to RHS:
$$ax^2 + bx = -c$$
3. Multiply the entire equation by $4a$ (Sreedharacharya's device):
$$4a(ax^2 + bx) = 4a(-c) \implies 4a^2x^2 + 4abx = -4ac$$
4. Recognize $4a^2x^2 = (2ax)^2$ and $4abx = 2(2ax)(b)$. Add $\mathbf{b^2}$ to both sides:
$$(2ax)^2 + 2(2ax)(b) + b^2 = b^2 - 4ac$$
5. Factor the LHS as a perfect square:
$$(2ax + b)^2 = b^2 - 4ac$$
6. Take the square root of both sides (provided $b^2 - 4ac \geq 0$):
$$2ax + b = \pm\sqrt{b^2 - 4ac}$$
7. Solve for $x$ by isolating $2ax$:
$$2ax = -b \pm \sqrt{b^2 - 4ac} \implies \mathbf{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

### Worked Archetype 4 | Application of the Quadratic Formula

**Problem:** Solve the equation $2x^2 - 7x + 3 = 0$ using the quadratic formula.

**Step 1: Identify coefficients:** Comparing with $ax^2 + bx + c = 0 \implies a = 2,\ b = -7,\ c = 3$.

**Step 2: Calculate the discriminant $D = b^2 - 4ac$:**

$$D = (-7)^2 - 4(2)(3) = 49 - 24 = \mathbf{25} > \mathbf{0} \implies \text{Two distinct real roots exist.}$$

**Step 3: Substitute into the quadratic formula:**

$$x = \frac{-(-7) \pm \sqrt{25}}{2(2)} = \frac{7 \pm 5}{4}$$

**Step 4: Separate the two roots:**

$$x_1 = \frac{7+5}{4} = \frac{12}{4} = \mathbf{3}, \qquad x_2 = \frac{7-5}{4} = \frac{2}{4} = \mathbf{\frac{1}{2}}$$

**Conclusion:** The roots are **3** and $\frac{1}{2}$.

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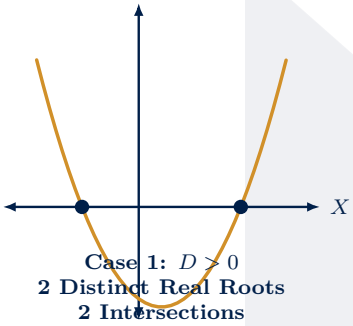
CORE SYLLABUS

# QUADRATIC EQUATIONS

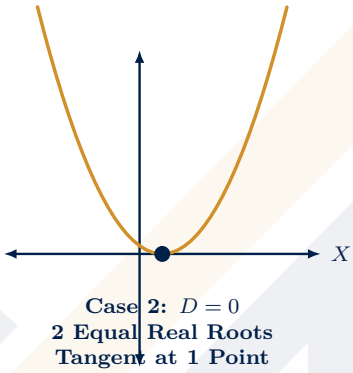
Module 5 | The Discriminant ($D$) & Tri-State Nature of Roots Matrix

## 5.1 The Discriminant Function $D = b^2 - 4ac$

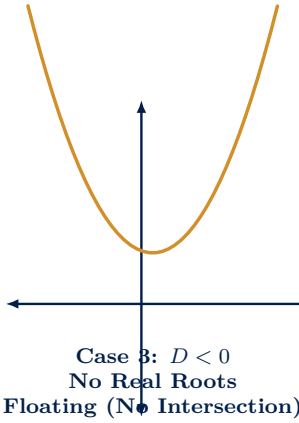
In the quadratic formula $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$, the quantity inside the square root, $\mathbf{D = b^2 - 4ac}$, is called the **discriminant** because it *discriminates* between the different types of roots.




**Case 1:** $D > 0$
**2 Distinct Real Roots**
**2 Intersections**




**Case 2:** $D = 0$
**2 Equal Real Roots**
**Tangent at 1 Point**



**Case 3:** $D < 0$
**No Real Roots**
**Floating (No Intersection)**

## 5.2 The Comprehensive Tri-State Nature of Roots Table

| Condition on $D$ | Value of $\sqrt{D}$ | Nature of Roots | Exact Formula for Roots |
|---|---|---|---|
| $\mathbf{D > 0}$ | Real & Positive | **Two distinct real roots** | $x = \frac{-b \pm \sqrt{D}}{2a}$ (Rational if $D$ is perfect square; conjugate surds if not). |
| $\mathbf{D = 0}$ | Zero (0) | **Two equal real roots** (co-incident) | $\mathbf{x = -\frac{b}{2a}}$, $-\frac{\mathbf{b}}{\mathbf{2a}}$ (Parabola touches $x$-axis at vertex). |
| $\mathbf{D < 0}$ | Imaginary | **No real roots** | No real values of $x$. Parabola does not touch $x$-axis. |

### Worked Archetype 5 | Inspecting the Nature of Roots

**Problem:** Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ and hence find the nature of its roots.
**Solution:** Given $2x^2 - 4x + 3 = 0 \implies a = 2,\ b = -4,\ c = 3$.

$$D = b^2 - 4ac = (-4)^2 - 4(2)(3) = 16 - 24 = \mathbf{-8}$$

Since $\mathbf{D < 0}$, the given quadratic equation has **no real roots**.

### Worked Archetype 6 | Coincident Roots ($D = 0$)

**Problem:** Find the nature of roots of $3x^2 - 2\sqrt{6}x + 2 = 0$. If real roots exist, find them.

$$D = b^2 - 4ac = (-2\sqrt{6})^2 - 4(3)(2) = [4 \times 6] - 24 = 24 - 24 = \mathbf{0}$$

Since $\mathbf{D = 0}$, the equation has **two equal real roots**:

$$x = -\frac{b}{2a} = -\frac{-2\sqrt{6}}{2(3)} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3} = \sqrt{\mathbf{\frac{2}{3}}}$$

Hence, both roots are $\sqrt{\frac{\mathbf{2}}{\mathbf{3}}}$ and $\sqrt{\frac{\mathbf{2}}{\mathbf{3}}}$.

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# QUADRATIC EQUATIONS

Module 6 | The Parameter $k$ Protocol: Condition for Equal Roots ($D = 0$)

## 6.1 The Tangency Condition and Parameter Determination

A quintessential question in the CBSE Class 10 board examination requires finding the value(s) of an unknown parameter $k$ such that the quadratic equation has **two equal real roots** (or coincident roots).

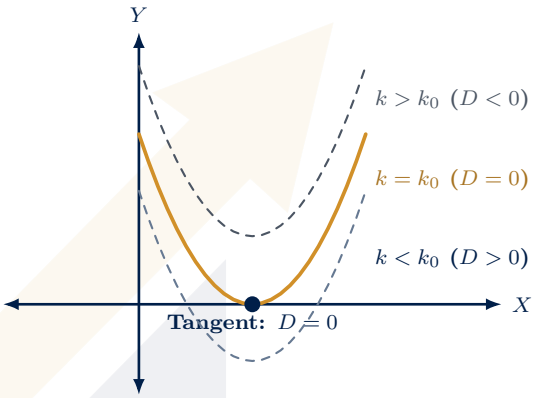
**The Dual Criteria for Equal Roots:**

1. **Criterion 1 (Discriminant Zero):**

$$\mathbf{D = 0 \iff b^2 - 4ac = 0 \iff b^2 = 4ac}$$

2. **Criterion 2 (Non-Zero Leading Coefficient):**

$$\mathbf{a \neq 0}$$




TRAP ALERT

**The Fatal Parameter Trap:** If the coefficient of $x^2$ contains $k$ (e.g., $a = k - 12$), you **must explicitly reject** any value of $k$ that makes $a = 0$!

## 6.2 Step-by-Step Equal Roots Solution Protocol

1. Expand all brackets and collect like terms to arrange the equation in strict standard form: $Ax^2 + Bx + C = 0$.
2. Write down the expressions for coefficients $a, b, c$ in terms of $k$.
3. Set up the discriminant equation: $D = b^2 - 4ac = 0$.
4. Solve the resulting algebraic equation for $k$.
5. **Mandatory Check:** Ensure that the values of $k$ do not cause the leading coefficient $a$ to become zero.

### Worked Archetype 7 | The Classic Leading Coefficient Trap

**Problem:** Find the value(s) of $k$ for which the quadratic equation $(k-12)x^2 + 2(k-12)x + 2 = 0$ has two equal real roots.

**Step 1: Identify coefficients:**

$$a = k - 12, \quad b = 2(k-12), \quad c = 2$$

**Constraint:** For the equation to remain quadratic, $a \neq 0 \implies k - 12 \neq 0 \implies \mathbf{k \neq 12}$.

**Step 2: Apply the equal roots condition $D = 0$:**

$$D = b^2 - 4ac = [2(k-12)]^2 - 4(k-12)(2) = 0$$

$$4(k-12)^2 - 8(k-12) = 0$$

**Step 3: Factor out the common term $4(k-12)$ (DO NOT divide by $k-12$ blindly!):**

$$4(k-12)\big[(k-12) - 2\big] = 0$$

$$4(k-12)(k-14) = 0$$

This yields two algebraic solutions:

$$k - 12 = 0 \implies k = 12 \quad \text{or} \quad k - 14 = 0 \implies k = 14$$

**Step 4: Check domain validity:** If $k = 12$, then $a = 12 - 12 = 0$, so the equation collapses to $0 \cdot x^2 + 0 \cdot x + 2 = 0 \iff 2 = 0$ (impossible!).

Hence, $\mathbf{k = 12}$ is **extraneous and must be rejected**.

**Conclusion:** The only valid value of $k$ is $\mathbf{k = 14}$.

CHAPTER 04

CORE SYLLABUS

# QUADRATIC EQUATIONS

Module 7 | Word Problems I: Numbers, Reciprocals & Age Models

## 7.1 Translation Protocol from Natural Language to Quadratic Equations

Word problems test your ability to translate verbal relationships into standard quadratic equations:

| Verbal Problem Model | Algebraic Representation | Standard Form Equation |
|---|---|---|
| Two consecutive integers | $x$ and $x+1$ | Product $= P \implies x(x+1) = P \implies x^2 + x - P = 0$ |
| Two consecutive odd/even integers | $x$ and $x+2$ | Sum of squares $= S \implies x^2 + (x+2)^2 = S$ |
| A number and its reciprocal | $x$ and $\frac{1}{x}$ $(x \neq 0)$ | $x + \frac{1}{x} = \frac{p}{q} \implies qx^2 - px + q = 0$ |
| Two numbers with sum $S$, product $P$ | $x$ and $S - x$ | $x(S-x) = P \implies x^2 - Sx + P = 0$ |
| Age problems (years ago / hence) | Present age $= x$; Past $= x - n$; Future $= x + m$ | Product of ages $= P \implies (x-n)(x+m) = P$ |

## 7.2 Domain Validation: Rejecting Non-Physical Solutions

TRAP ALERT

**Physical Variables Have Strict Domain Restrictions:** Quadratic equations mathematically yield two roots. In word problems, you must test both roots against physical reality:

- **Ages:** Must be positive real numbers ($x > 0$). Reject negative roots!
- **Natural Numbers / Counts:** Must be positive integers ($x \in \mathbb{N}$).
- **Dimensions (Length, Width):** Must be strictly positive ($x > 0$).

### Worked Archetype 8 | The Reciprocal Age Problem

**Problem:** The sum of the reciprocals of Rehman's ages (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age.

**Step 1: Define the variable:** Let Rehman's present age be $x$ years ($x > 3$).
Rehman's age 3 years ago $= (x - 3)$ years.
Rehman's age 5 years from now $= (x + 5)$ years.

**Step 2: Formulate the equation according to given conditions:**

$$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3}$$

**Step 3: Combine fractions by taking LCM on LHS:**

$$\frac{(x+5)+(x-3)}{(x-3)(x+5)} = \frac{1}{3} \implies \frac{2x+2}{x^2+2x-15} = \frac{1}{3}$$

**Step 4: Cross-multiply and simplify to standard form:**

$$3(2x+2) = x^2 + 2x - 15 \implies 6x + 6 = x^2 + 2x - 15$$

$$x^2 + 2x - 6x - 15 - 6 = 0 \implies \mathbf{x^2 - 4x - 21 = 0}$$

**Step 5: Factor by splitting the middle term ($ac = -21, b = -4 \implies -7$ and $3$):**

$$(x-7)(x+3) = 0 \implies x = 7 \quad \text{or} \quad x = -3$$

**Step 6: Physical validation:** Age cannot be negative, so $x = -3$ is rejected.
**Conclusion:** Rehman's present age is **7** years.

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# QUADRATIC EQUATIONS

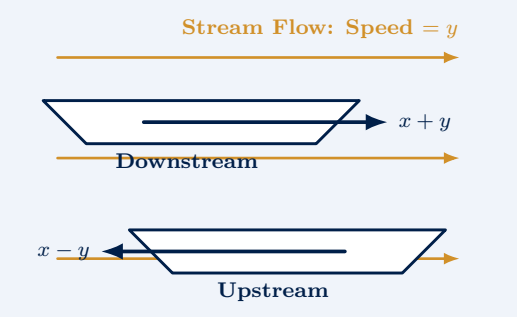
Module 8 | Word Problems II: Speed, Distance & River Kinematics

## 8.1 The Speed-Distance-Time Identity: $t = \frac{d}{s}$

All uniform motion problems rely on the relation Time $= \frac{\text{Distance}}{\text{Speed}}$. When speed changes, a time differential $\Delta t$ is created between the slow and fast trips:

$$\mathbf{t_{slow} - t_{fast} = \Delta t \implies \frac{d}{s} - \frac{d}{s+v} = \Delta t}$$

## 8.2 River Stream Kinematics (Upstream vs. Downstream)




**Relative Velocities on Water:** Let boat speed in still water be $x$ km/h, and stream speed be $y$ km/h ($x > y$):

- **Downstream (with flow):**

$$\mathbf{v_{down} = x + y}$$

- **Upstream (against flow):**

$$\mathbf{v_{up} = x - y}$$

- **Time Equation for Distance $d$:** Since $x - y < x + y$, upstream travel takes **longer**:

$$\mathbf{t_{up} - t_{down} = \Delta t \iff \frac{d}{x-y} - \frac{d}{x+y} = \Delta t}$$

### Worked Archetype 9 | The Motorboat Stream Problem

**Problem:** A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

**Step 1: Set up variables:** Let the speed of the stream be $y$ km/h ($0 < y < 18$).
Speed upstream $= (18 - y)$ km/h, Speed downstream $= (18 + y)$ km/h.

**Step 2: Formulate the time equation ($t_{up} - t_{down} = 1$ hour):**

$$\frac{24}{18-y} - \frac{24}{18+y} = 1$$

**Step 3: Factor out 24 and combine fractions:**

$$24\left[\frac{(18+y)-(18-y)}{(18-y)(18+y)}\right] = 1 \implies 24\left[\frac{2y}{324-y^2}\right] = 1$$

$$\frac{48y}{324-y^2} = 1 \implies 48y = 324 - y^2 \implies \mathbf{y^2 + 48y - 324 = 0}$$

**Step 4: Factor the quadratic equation ($ac = -324, b = 48 \implies 54$ and $-6$):**

$$(y + 54)(y - 6) = 0 \implies y = -54 \quad \text{or} \quad y = 6$$

**Step 5: Physical validation:** Speed cannot be negative $\implies y = -54$ is rejected.
**Conclusion:** The speed of the stream is **6** km/h.

CHAPTER 04

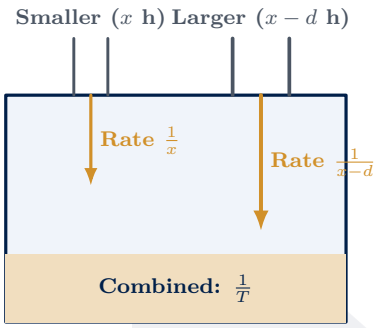
CORE SYLLABUS

# QUADRATIC EQUATIONS

Module 9 | Word Problems III: Water Taps, Cisterns & Unit Work Rates

## 10.1 The Unit Work Rate Principle

If a pipe or worker can complete a task in $T$ hours, then in 1 hour, the fraction of work completed is exactly $\frac{1}{\mathbf{T}}$. When two pipes work simultaneously, their individual flow rates add up to equal the combined flow rate:




**Master Tap Flow Rate Equation:** Let smaller tap take $x$ hours, and larger tap take $(x-d)$ hours to fill tank alone:

$$\text{Rate of Tap 1} = \frac{\mathbf{1}}{\mathbf{x}}$$

$$\text{Rate of Tap 2} = \frac{\mathbf{1}}{\mathbf{x-d}}$$

$$\text{Combined Rate} = \frac{\mathbf{1}}{\mathbf{x}} + \frac{\mathbf{1}}{\mathbf{x-d}} = \frac{\mathbf{1}}{\mathbf{T}_{\text{together}}}$$

*Clearing denominators produces a clean quadratic equation in $x$!*

### Worked Archetype 10 | Two Water Taps Filling a Tank Together

**Problem:** Two water taps together can fill a tank in $9\frac{3}{8}$ hours ($= \frac{75}{8}$ hours). The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

**Step 1: Define variables:** Let time taken by smaller tap alone $= x$ hours ($x > 10$).
Time taken by larger tap alone $= (x-10)$ hours.
Combined time $T = \frac{75}{8}$ hours $\implies$ Combined 1-hour rate $= \frac{1}{T} = \frac{8}{75}$.

**Step 2: Set up the flow rate equation:**

$$\frac{1}{x} + \frac{1}{x-10} = \frac{8}{75}$$

**Step 3: Combine LHS and cross-multiply:**

$$\frac{(x-10)+x}{x(x-10)} = \frac{8}{75} \implies \frac{2x-10}{x^2-10x} = \frac{8}{75}$$

Divide numerator of LHS and RHS by 2:

$$\frac{x-5}{x^2-10x} = \frac{4}{75} \implies 75(x-5) = 4(x^2-10x)$$

$$75x - 375 = 4x^2 - 40x \implies \mathbf{4x^2 - 115x + 375 = 0}$$

**Step 4: Solve the quadratic equation by factoring ($ac = 4 \times 375 = 1500, b = -115 \implies -100$ and $-15$):**

$$4x^2 - 100x - 15x + 375 = 0 \implies 4x(x-25) - 15(x-25) = 0$$

$$(x-25)(4x-15) = 0 \implies x = 25 \quad \text{or} \quad x = \frac{15}{4} = 3.75$$

**Step 5: Physical validity check:** If $x = 3.75$ hours, then time for larger tap $= 3.75 - 10 = -6.25$ hours (impossible!).
Hence, $x = 3.75$ is **rejected**, and we accept $\mathbf{x = 25}$.
**Conclusion:** Time for smaller tap alone is **25** hours, and for larger tap alone is **15** hours.

CHAPTER 04

CORE SYLLABUS

# QUADRATIC EQUATIONS

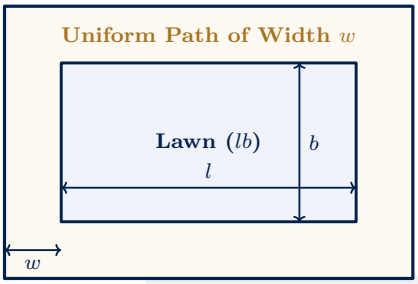
Module 10 | Word Problems IV: Geometry, Mensuration & Pathways

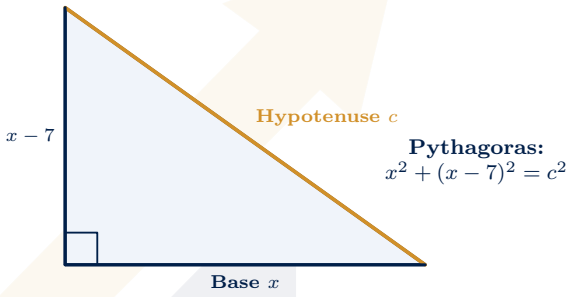
## 11.1 Geometric Formulations via Pythagoras & Area Theorems

Mensuration word problems translate spatial configurations into quadratic forms:

- **Right Triangles:** Hypotenuse $c$, sides $x$ and $x-d \implies \mathbf{x^2 + (x-d)^2 = c^2}$.
- **Rectangular Fields with Uniform Borders:** An inner lawn $(l \times b)$ surrounded by a path of width $w$:

$$\text{Outer Length} = l + 2w, \quad \text{Outer Breadth} = b + 2w \implies \text{Area of Path} = \mathbf{(l+2w)(b+2w) - lb}$$




### Worked Archetype 11 | Right Triangle Altitude and Base Relationship

**Problem:** The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

**Step 1: Formulate variables:** Let base $= x$ cm $(x > 7)$. Altitude $= (x-7)$ cm. Hypotenuse $= 13$ cm.

**Step 2: Apply Pythagoras Theorem:**

$$\text{Base}^2 + \text{Altitude}^2 = \text{Hypotenuse}^2 \implies x^2 + (x-7)^2 = 13^2$$

**Step 3: Expand and simplify to standard form:**

$$x^2 + (x^2 - 14x + 49) = 169 \implies 2x^2 - 14x + 49 - 169 = 0$$

$$2x^2 - 14x - 120 = 0 \implies \text{Divide by 2: } \mathbf{x^2 - 7x - 60 = 0}$$

**Step 4: Factor the quadratic equation** ($ac = -60, b = -7 \implies -12$ **and** $5$**):**

$$(x-12)(x+5) = 0 \implies x = 12 \quad \text{or} \quad x = -5$$

**Step 5: Physical validation:** Side length cannot be negative $\implies x = -5$ is rejected.

$$\text{Base } x = \mathbf{12} \text{ cm}, \quad \text{Altitude } = x - 7 = 12 - 7 = \mathbf{5} \text{ cm}$$

**Verification:** $12^2 + 5^2 = 144 + 25 = 169 = 13^2$ (Pythagorean triplet verified!).

TUITION CLASSES

CHAPTER 04

QUICK REFERENCE

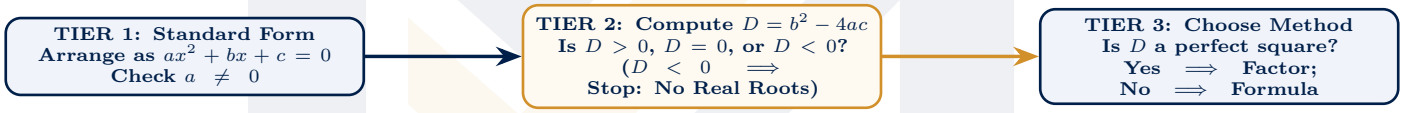
# QUADRATIC EQUATIONS

Module 11 | Master Formula Matrix & Solution Decision Protocol

## 11.1 The Comprehensive Quadratic Equations Matrix

| Method / Theorem | Prerequisite Form | Mathematical Rule | Strategic Board Examination Purpose |
|---|---|---|---|
| **Standard Form** | $a \neq 0$ | $ax^2 + bx + c = 0$ | Mandatory first step for all quadratic methods. |
| **Factorization** | $p \cdot q = ac,\ p + q = b$ | $(px + r)(qx + s) = 0$ | Fastest method when $ac$ factors cleanly. |
| **Quadratic Formula** | $D \geq 0$ | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ | Universal method; handles surd/radical coefficients. |
| **Discriminant** $D$ | $D = b^2 - 4ac$ | $D > 0,\ D = 0,\ D < 0$ | Testing nature of roots without solving equation. |
| **Equal Roots Condition** | Two equal roots | $b^2 - 4ac = 0$ with $a \neq 0$ | Finding unknown parameter $k$ (high frequency!). |
| **Parabola Vertex** | Axis of symmetry | $x = -\frac{b}{2a},\ y = -\frac{D}{4a}$ | Extreme values / trajectory height peak. |
| **Sum & Product** | Roots $\alpha, \beta$ | $\alpha + \beta = -\frac{b}{a},\ \alpha\beta = \frac{c}{a}$ | Fast verification check of computed roots. |

## 11.2 The 3-Tier Board Problem Decision Flowchart




**Fast Mental-Math Factoring Rules:**

- **Sign of $c$ is Positive ($c > 0$):** Both factors must have the **same sign** (the sign of $b$).
  - $x^2 - 8x + 12 = 0 \implies b < 0, c > 0 \implies$ both negative: $(x - 6)(x - 2) = 0$.
  - $x^2 + 8x + 12 = 0 \implies b > 0, c > 0 \implies$ both positive: $(x + 6)(x + 2) = 0$.
- **Sign of $c$ is Negative ($c < 0$):** Factors have **opposite signs**; the larger factor takes the sign of $b$.
  - $x^2 - x - 12 = 0 \implies b < 0, c < 0 \implies$ larger factor is negative: $(x - 4)(x + 3) = 0$.

MATHS MANTHAN TUITION CLASSES

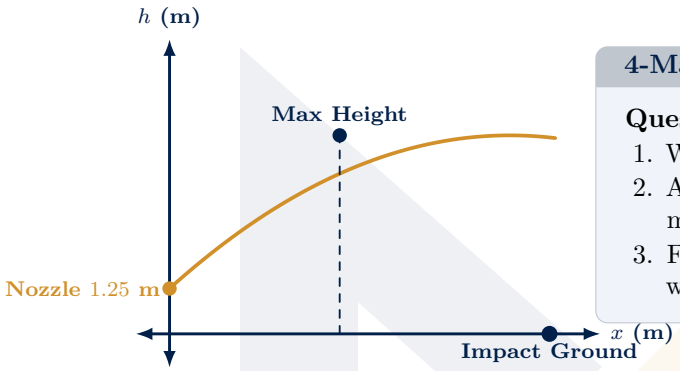
# QUADRATIC EQUATIONS

Module 12 | Applied Case-Study: Projectile Motion & Parabolic Flight

## Case Study 1 | Fire Service Water Cannon Trajectory

A specialized fire-fighting truck sprays water through a pressurized cannon. The height $h$ (in meters) of the water jet above the ground at a horizontal distance $x$ (in meters) from the nozzle is modeled by the quadratic equation:

$$h(x) = -\frac{1}{20}x^2 + x + 1.25$$




### 4-Mark Board Question Protocol

**Questions:**

1. What is the height of the nozzle from the ground? [1M]
2. At what horizontal distance does the water jet reach its maximum height? [1M]
3. Find the horizontal distance from the nozzle where the water jet strikes the ground. [2M]

### Solution & Marking Scheme Evaluation

1. **Height of Nozzle [1M]:** At the nozzle, the horizontal distance is $x = 0$:

$$h(0) = -\frac{1}{20}(0)^2 + (0) + 1.25 = \mathbf{1.25} \text{ meters}$$

2. **Distance for Maximum Height [1M]:** Maximum height occurs at the vertex $x = -\frac{b}{2a}$:

$$a = -\frac{1}{20} = -0.05, \quad b = 1 \implies x = -\frac{1}{2(-0.05)} = -\frac{1}{-0.1} = \mathbf{10} \text{ meters}$$

3. **Impact Horizontal Distance [2M]:** Water strikes ground when height $h(x) = 0$:

$$-\frac{1}{20}x^2 + x + 1.25 = 0 \implies \text{Multiply by } -20: \quad \mathbf{x^2 - 20x - 25 = 0}$$

Apply the quadratic formula with $a = 1, b = -20, c = -25$:

$$D = (-20)^2 - 4(1)(-25) = 400 + 100 = 500$$

$$x = \frac{-(-20) \pm \sqrt{500}}{2(1)} = \frac{20 \pm 10\sqrt{5}}{2} = 10 \pm 5\sqrt{5}$$

Since distance must be positive, take the positive root ($\sqrt{5} \approx 2.236$):

$$x = 10 + 5(2.236) = 10 + 11.18 = \mathbf{21.18} \text{ meters}$$

CHAPTER 04

EXAM STRATEGY

# QUADRATIC EQUATIONS

Module 13 | Board Traps, Verification Hacks & 5-Point Audit

## 13.1 The Top 7 Quadratic Equation Board Traps (And How to Bust Them)

**Trap 1: The Variable Division Cancellation Trap:** Solving $x^2 = 5x$ by dividing both sides by $x$ to get $x = 5$.

TRAP ALERT

You lose the valid root $\mathbf{x = 0}$! Always factor: $x^2 - 5x = 0 \implies x(x-5) = 0 \implies x = 0, 5$.

**Trap 2: The Negative Linear Coefficient Sign Error:** In $x = \frac{-b \pm \sqrt{D}}{2a}$, when $b = -7$, writing $-7 \pm \ldots$ instead of $-(-7) = +\mathbf{7}$.

**Trap 3: The Squaring Negative $b$ Blunder:** Writing $(-6)^2 = -36$ inside $b^2 - 4ac$. Any real number squared is non-negative: $\mathbf{(-6)^2 = +36}$!

**Trap 4: The Parameter $a \neq 0$ Violation:** In $(k-1)x^2 + 2x + 1 = 0$, finding $k = 1$ from $D = 0$ but forgetting that $k = 1$ makes $a = 0$. You **must reject** values making the leading coefficient zero!

**Trap 5: Upstream vs. Downstream Subtraction Order:** Writing downstream time minus upstream time as positive. Upstream is always **slower**, so $\mathbf{t}_{\text{upstream}} > \mathbf{t}_{\text{downstream}}$.

**Trap 6: Retaining Extraneous Negative Roots in Word Problems:** Forgetting to state: "Since speed/age/time cannot be negative, $x = -k$ is rejected."

**Trap 7: Using the Deleted "Completing the Square" Method:** In CBSE 2026–27, solve strictly using **Factorization** or the **Quadratic Formula**.

## 13.2 High-Speed Algebraic Verification Protocol (Vieta's Relations)

After calculating roots $\alpha$ and $\beta$, check them mentally in 5 seconds using Vieta's formulas:

$$\text{Sum of Roots: } \alpha + \beta = -\frac{\mathbf{b}}{\mathbf{a}} \qquad \text{and} \qquad \text{Product of Roots: } \alpha \cdot \beta = \frac{\mathbf{c}}{\mathbf{a}}$$

If the sum or product does not match the original equation coefficients, an arithmetic error occurred!

## 13.3 The 5-Point Board Examination Pre-Submission Checklist

**Final 60-Second Answer Sheet Verification Protocol:**

- ☐ **Standard Form Established:** Did you gather all terms to one side ($ax^2 + bx + c = 0$) before identifying $a, b, c$?
- ☐ **Discriminant Sign Verified:** Did you compute $D = b^2 - 4ac$ accurately with correct sign multiplication $(-4ac)$?
- ☐ **Two Roots Recorded:** Did you state both roots explicitly? (Quadratic equations have 2 roots!).
- ☐ **Physical Feasibility Justified:** For word problems, did you write an explicit sentence rejecting the negative root?
- ☐ **Correct Dimensional Units Attached:** Speeds in km/h, Times in hours, Dimensions in cm/m!

**MATHS MANTHAN TUITION CLASSES | END OF CHAPTER 4 HANDBOOK**

Designed for Academic Excellence in CBSE Class 10 Board Examinations 2026–27.